

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

4. Modulation–Demodulation Methods in Mobile Communications

4 hrs · asked in **20 of 22 papers** · 5 syllabus sub-topics · **no paper has ever set a numerical here** (one worked example is given anyway, §4.4).

Read this first:

- **This is a drawing chapter.** Almost every question says +Fig, block diagram or constellation diagram. Marks come from the **diagram**, not the prose.
- Four blocks carry the chapter, in order of return:
 - **OFDM** **TOP 9/22** → transmitter + receiver block diagram, always 4 or 8 marks.
 - **Spread spectrum, DSSS and FHSS** **TOP 7/22** → definition + block diagram.
 - **MSK / GMSK** **TOP 6/22** → why GSM picked GMSK.
 - **QPSK** **HOT 4/22** → transmitter, receiver, constellation.
- Everything else (BPSK, DPSK, OQPSK, $\pi/4$ -QPSK, QAM) is a **one-paper** question, but each was worth 6 to 8 marks when it came.
- **Learn 3 diagrams cold:** QPSK transmitter/receiver pair, OFDM transmitter/receiver pair, DSSS + FHSS transmitter pair. They answer roughly 30 marks of the paper.
- FHSS appears here as a *modulation* and again in Ch 7 as **FHMA**, a *multiple access* method. Same block diagram, different framing. Draw it once, label it for whichever chapter is asked.

4.1 Why modulate, why digital, and how schemes are judged

Only one PYQ lives in this section, and it is worth 2 marks. Everything else here is **vocabulary the rest of the chapter assumes**: η_p , η_B , the trade-off between them, and what a constellation diagram actually plots. Skim it once, then never look at it again.

Advantages of digital modulation over analog 2

79 Bh 2 · always a rider on the GMSK question, never asked alone

Modulation itself

- Process of impressing message on high-freq carrier suitable for transmission.
- Translates baseband (modulating) signal $\rightarrow$ bandpass (modulated) signal.
- **Why:** antenna size $\propto \lambda$, so baseband speech at 3 kHz needs km-long antennas.
- **Why:** shifting to different carriers lets many users share one medium (multiple access).
- **Why:** higher freq $\rightarrow$ longer range, smaller hardware.

Power efficiency η_p

- Ability to hold message quality at low power (low SNR).
- $\eta_p = E_b/N_0$ at a stated BER, usually 10^{-5} .
- E_b = energy per bit, N_0 = noise power spectral density.
- Trade-off: BER $\uparrow$ as $E_b/N_0 \downarrow$.
- **Lower E_b/N_0 for the same BER = more power efficient.**

Choosing a scheme for a mobile link

- Low BER at low SNR (handset battery is finite).
- Resistance to interference (ACI, CCI) and multipath fading.
- Minimum occupied bandwidth (spectrum is bought, not free).
- Cheap, small, low-power to implement in the handset.
- **No scheme meets all four**, so every standard is a trade-off.

Constellation (signal space) diagram

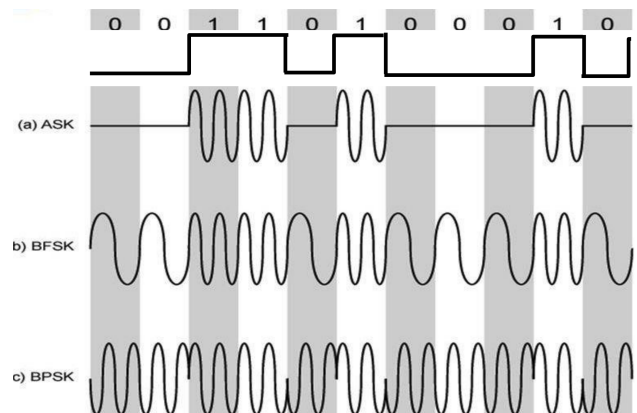
- Graphical plot of the complex envelope of every possible symbol.
- x axis = in-phase component, y axis = quadrature component.
- Distance of a point from origin = amplitude, so distance² = symbol power.
- **Distance d between the two nearest points sets noise immunity.**
- $P_e \propto Q\left(\frac{d}{\sqrt{2N_0}}\right)$, so small $d \rightarrow$ high BER.
- Farthest point from origin = peak power the amplifier must deliver.

Digital over analog 2

- Regenerable: noise does not accumulate, receiver rebuilds clean symbols.
- Resistant to fading, noise, interference.
- Voice + data + video multiplexed on one channel.
- Encryption possible $\rightarrow$ security and privacy.
- Error control coding detects and corrects errors (Ch 6).
- Built in DSP software, not analog hardware $\rightarrow$ cheap, small, reprogrammable.
- Equalization and diversity apply cleanly (Ch 5).
- **Cost:** needs more bandwidth per message and needs synchronisation.

Bandwidth efficiency η_B

- Ability to carry data in a limited band.
- $\eta_B = R/B$ bps/Hz, R = data rate, B = RF bandwidth.
- Ceiling set by Shannon:
 $\eta_{B(\max)} = C/B = \log_2(1 + S/N)$.
- **The fundamental trade-off:** improve η_p and η_B gets worse, and the reverse. No scheme wins both.
- M-ary buys η_B by spending η_p .



The three binary keying waveforms against one data stream: (a) ASK, (b) BFSK, (c) BPSK. SST, Modulation extra note

4.2 Review: analog modulation and line coding syllabus, no PYQ yet

Amplitude family

- **AM (DSB-FC):** $B = 2f_m$. Carrier carries no information, so at most 33 % of power is useful. Simple envelope detector.
- **DSB-SC:** carrier suppressed, $B = 2f_m$. All power in sidebands, but needs coherent detection.
- **SSB:** one sideband only, $B = f_m$. **Half the bandwidth, half the power**, but sharp filters and exact carrier recovery needed.
- **VSF:** one full sideband + vestige of the other, $B \approx 1.25f_m$. Compromise used for TV video where DC content forbids true SSB.

Line coding: what it is for

- Maps bits onto baseband pulses before modulation.
- Requirements: no DC content, self-clocking, spectrally compact, error detection possible, transparent to any bit pattern.

Angle family

- **FM:** instantaneous freq varies with message. Carson's rule $B = 2(\Delta f + f_m)$.
- Constant envelope → non-linear class-C amplifier usable → power efficient.
- **Capture effect:** stronger of two co-channel FM signals suppresses the weaker.
- Wideband FM trades bandwidth for SNR → why **every 1G system used FM**.
- **PM:** phase varies with message. FM and PM are inter-convertible through an integrator or differentiator.

The four to know

- **NRZ:** 1 = high for full bit, 0 = low. Compact, but has DC and no clock in long runs. **GSM uses bipolar NRZ ahead of the GMSK filter.**
- **RZ:** pulse returns to zero mid-bit. Self-clocking, costs twice the bandwidth.
- **Manchester:** transition mid-bit, 1 = hi→lo, 0 = lo→hi. No DC, always self-clocking, twice the bandwidth. Used in 10 Mbps Ethernet.
- **AMI (bipolar):** 0 = zero level, 1 = alternating $\pm$. No DC, and a run of same-polarity ones flags an error.

4.3 Linear modulation: BPSK, DPSK, QPSK, OQPSK, $\pi/4$ -QPSK

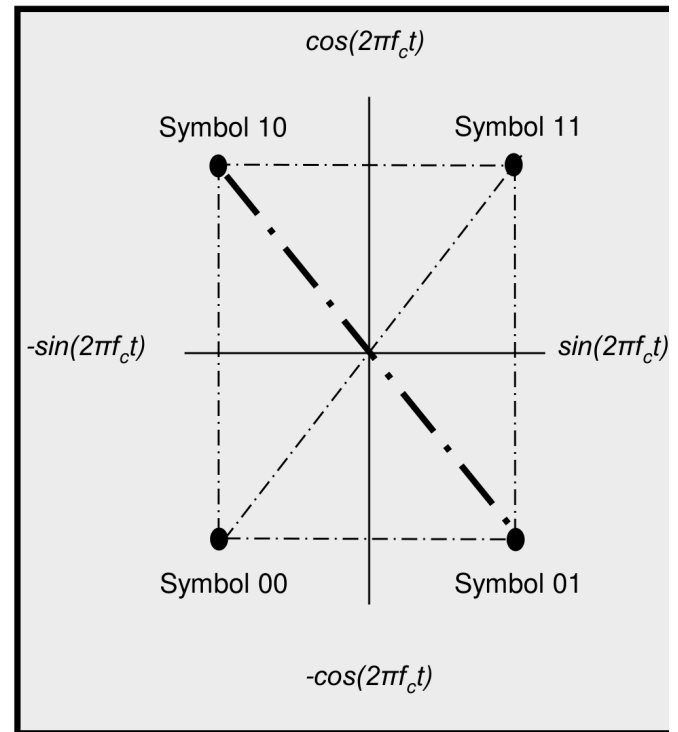
Linear means the transmitted amplitude follows the modulating signal linearly. Bandwidth efficient, but envelope varies → **needs a linear (backed-off, inefficient) power amplifier**. That is exactly why mobile handsets moved to the constant-envelope family in §4.4.

QPSK: transmission and detection process + Fig, constellation diagram 6 7 8

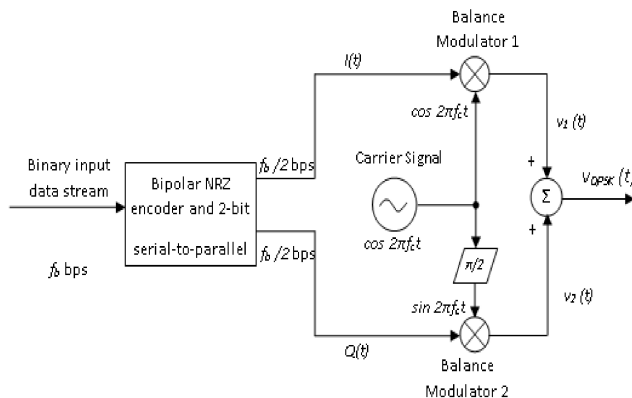
HOT 4/22 75 Bh 6, 72 Ma 8, 81 Bh 8, 73 Bh 7 · answer = two block diagrams + one constellation + six lines

Idea

- M-ary PSK with $M = 4$, so **2 bits per symbol** (a dibit).
- Carrier phase takes one of 4 equally spaced values $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.
- $s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + (2i - 1)\frac{\pi}{4})$, $i = 1, 2, 3, 4$.
- Symbol period $T_s = 2T_b \rightarrow$ **half the bandwidth of BPSK for the same bit rate**.
- Null-to-null RF bandwidth $= R_b$ (BPSK needs $2R_b$), so $\eta_B = 1$ bps/Hz.
- **BER identical to BPSK**: the dibit splits into two independent BPSK streams on orthogonal carriers, $P_e = Q\left(\sqrt{2E_b/N_0}\right)$.
- So QPSK gets double the bandwidth efficiency **for free**. That is the whole selling point.
- Symbols are **Gray coded** (00, 01, 11, 10 around the circle) so a wrong symbol costs only 1 bit.



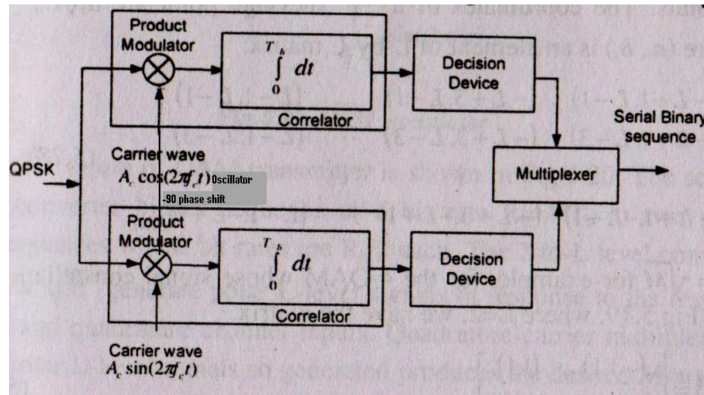
QPSK constellation: 4 points, one per dibit, spaced 90° . Distance to origin $= \sqrt{E_s}$, distance between neighbours $= \sqrt{2E_s} = 2\sqrt{E_b}$. SST, Modulation extra note



QPSK transmitter. SST, Fig 3.2.3

Transmission process 4

1. Binary stream at f_b bps enters a **bipolar NRZ encoder** $\rightarrow \pm 1$ levels.
2. **Serial-to-parallel converter** splits it: even-numbered bits $\rightarrow I(t)$, odd-numbered bits $\rightarrow Q(t)$, each at $f_b/2$ bps.
3. $I(t)$ multiplied by $\cos 2\pi f_c t$ in balance modulator 1.
4. $Q(t)$ multiplied by $\sin 2\pi f_c t$ (carrier through a 90° shifter) in balance modulator 2.
5. **Summer** adds the two $\rightarrow$ QPSK signal, one of 4 phases per dibit.
6. Output BPF limits the spectrum to the allocated channel.



QPSK receiver, a pair of coherent correlators. SST, Modulation extra note

Detection process 4

1. Received QPSK splits into two arms (coherent detection, so carrier recovery circuit supplies $\cos$ and $\sin$ references).
2. Upper arm: product modulator with $\cos 2\pi f_c t$, then **integrate-and-dump** over 0 to $T_b \rightarrow$ output x_1 .
3. Lower arm: product modulator with $\sin 2\pi f_c t$, then integrator $\rightarrow$ output x_2 .
4. **Decision device**, threshold 0 V: $x_1 > 0 \rightarrow 1$, $x_1 < 0 \rightarrow 0$ on the I channel, same rule for x_2 on the Q channel.
5. **Multiplexer** interleaves the two recovered streams $\rightarrow$ original binary sequence.

Say "coherent" out loud. QPSK cannot be envelope detected, the receiver needs the carrier phase.

The weakness that produced OQPSK and $\pi/4$ -QPSK

- When *both* bits of a dibit change, the phase jumps 180° and the **envelope passes through zero**.
- After the transmitter's band-limiting filter this zero crossing becomes a deep amplitude dip.
- A non-linear (efficient) amplifier hard-limits that dip $\rightarrow$ the filtered sidelobes **grow back** $\rightarrow$ adjacent channel interference returns.
- Fixing this without giving up bandwidth efficiency is the whole story of §4.3 and §4.4.

BPSK against QPSK 5

73 Ma 5

Parameter	BPSK	QPSK
Bits per symbol	1	2 (dibit)
Symbol period	$T_s = T_b$	$T_s = 2T_b$
Constellation	2 points, 180° apart	4 points, 90° apart
Null-to-null RF BW	$2R_b$	R_b
η_B	0.5 bps/Hz	1 bps/Hz
Bit error rate	$Q(\sqrt{2E_b/N_0})$	$Q(\sqrt{2E_b/N_0})$, the same
Phase jumps	180° only	$0^\circ, \pm 90^\circ$ or 180°
Circuit	one balance modulator	two, plus S/P and 90° shifter

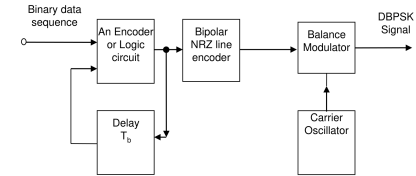
The one-line answer: QPSK carries twice the data in the same band at the same BER and the same E_b/N_0 , paying only in circuit complexity.

DPSK modulation: transmitter and receiver; PN sequence and why it is used 4+2+2

PIN 1/22 72 Ma 4+2+2 · the PN half of this question is really \$4.7 material, cross-refer it

What DPSK is

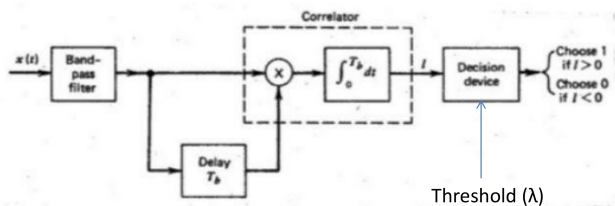
- **Non-coherent form of PSK**: no synchronous carrier needed at the receiver.
- Information sits in the **difference between the phases of two successive symbols**, not in the absolute phase.
- Encoding rule: symbol **1** = no phase transition, symbol **0** = 180° transition.
- So the present bit is decoded by comparing against the previous bit.
- $P_e = \frac{1}{2}e^{-E_b/N_0}$, about **3 dB worse** than coherent BPSK at high SNR.



DPSK transmitter: encoder + T_b delay ahead of an ordinary BPSK modulator. SST, Modulation extra note

Transmitter 2

1. Binary sequence m_k enters a **logic (differential encoder)** circuit.
2. A **1-bit delay** T_b feeds the previous encoded bit d_{k-1} back to it.
3. Encoder output d_k toggles when $m_k = 0$ and holds when $m_k = 1$.
4. **Bipolar NRZ encoder** $\rightarrow \pm 1$ levels.
5. **Balance modulator** multiplies by the carrier $\rightarrow$ DPSK out.



DPSK receiver: the signal is its own phase reference. SST, Modulation extra note

Receiver 2

1. **Bandpass filter** centred on f_c removes out-of-band noise.
2. Signal is multiplied by **its own T_b -delayed copy** in the correlator.
3. **Integrate and dump** over one bit gives an output proportional to $\cos \phi$, where ϕ is the phase difference between the two.
4. $\phi = 0 \rightarrow$ output positive $\rightarrow$ decide **1**.
 $\phi = \pi \rightarrow$ output negative $\rightarrow$ decide **0**.
5. **Decision device** at threshold $\lambda = 0$ V rebuilds the sequence.

No carrier recovery circuit anywhere in this diagram. That is the mark the examiner is looking for.

Binary data b(t)	0	0	1	0	0	1	0	0	1	1
Diff. encoded data d(k)	1*	0	1	1	0	1	1	0	1	1
Phase of DPSK	0	π	0	0	π	0	0	π	0	0
Shift Diff. encoded data d(k-1)	1	0	1	1	0	1	1	0	1	1
Phase of shift DPSK	0	π	0	0	π	0	0	π	0	0
Phase com. o/p	-	-	+	-	-	+	-	-	+	+
Detected Binary seq	0	0	1	0	0	1	0	0	1	1

Differential encoding worked through: the starred bit is the arbitrary reference. Row 2 is d_k , row 3 its phase, last row the recovered data. SST, Modulation extra note

DPSK against BPSK

- **Bandwidth** f_b against $2f_b \rightarrow$ DPSK halves it.
- **Carrier at demodulator:** not required against required.
- **Error probability:** higher against low.
- **Bit detection:** over two successive bit intervals against a single interval.
- **Circuit:** simpler receiver, but the encoder adds transmitter complexity.

(+) and (-)

- (+) No carrier recovery $\rightarrow$ cheap, small, fast to acquire.
- (+) Works where the channel rotates phase faster than a PLL can track.
- (-) P_e roughly 3 dB worse than coherent BPSK.
- (-) **Errors propagate in pairs:** one bad symbol corrupts the reference for the next.

PN sequence and why it is used 2+2 (full treatment in §4.7)

- **PN (pseudo-noise) sequence** = a coded sequence of 1s and 0s that looks random to anyone without the generator, but is **completely deterministic and repeatable**.
- Generated by an m -stage shift register with XOR (parity) feedback; maximal-length period = $2^m - 1$ chips.
- **Properties:** 1s and 0s roughly equiprobable; sharp autocorrelation peak and near-zero elsewhere; low cross-correlation between different codes; adding a shifted copy gives the same sequence in a new phase; easy to generate and synchronise.
- **Why used:** spreads the spectrum, so the signal hides under the noise floor (low probability of intercept) and shrugs off narrowband jamming.
- **Why used:** nearly orthogonal codes let many users share one band at once $\rightarrow$ CDMA.
- **Why used:** gives the receiver a unique pattern to correlate against, which is both the addressing scheme and the synchronisation reference.

OQPSK transmitter and receiver; why $\pi/4$ -QPSK is preferred over OQPSK 5+2

PIN 1/22 71 Bh 5+2

Offset QPSK (OQPSK), also called staggered QPSK

- Same circuit as QPSK, with one change: the Q arm is delayed by $T_b = T_s/2$ before its balance modulator.
- Effect: the even and odd streams can no longer change at the same instant.
- So only one of I or Q switches at a time $\rightarrow$ maximum phase change is $\pm 90^\circ$, never 180° .
- The envelope therefore never passes through zero, and band-limiting produces only shallow dips instead of nulls.
- Hard-limiting or class-C amplification no longer regenerates the sidelobes, so OQPSK keeps its spectral occupancy in a real transmitter.
- Phase transitions happen every T_b instead of every T_s , i.e. twice as often but half as violent.
- PSD, BER and bandwidth are identical to QPSK. The gain is purely in envelope behaviour.

Transmitter 2.5

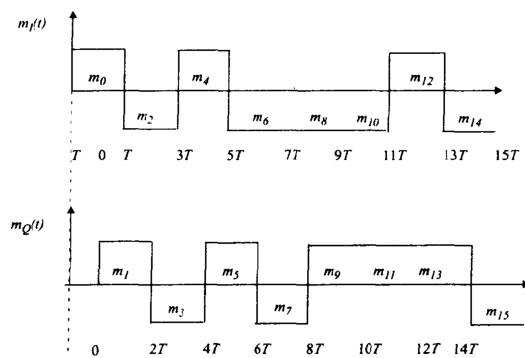
1. NRZ $\rightarrow$ serial-to-parallel into $m_I(t)$ and $m_Q(t)$.
2. Insert a T_b delay in the Q branch.
3. $m_I \times \cos 2\pi f_c t$, delayed $m_Q \times \sin 2\pi f_c t$.
4. Sum, then BPF $\rightarrow$ OQPSK out.

Receiver 2.5

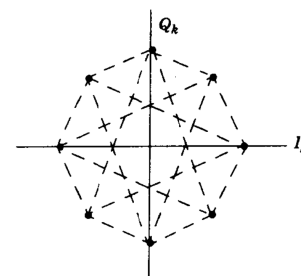
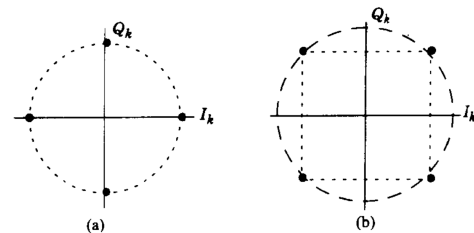
1. Coherent, exactly the QPSK receiver: BPF, carrier recovery, two correlators, two decision circuits.
2. The T_b stagger is removed by delaying the I decision (or advancing Q) before the multiplexer, so the dibits re-align.
3. Symbol timing recovery is taken from the transitions, which now occur every T_b .

$\pi/4$ -QPSK, the compromise

- Two QPSK constellations, offset from each other by $\pi/4$, used on alternate symbols.
- Signalling points therefore take 8 possible phase states, but only 4 are available in any one symbol.
- Maximum phase change is limited to $\pm 135^\circ$: worse than OQPSK's 90° , much better than QPSK's 180° .
- Every symbol carries a phase transition (no 0° option), which makes clock recovery easy.
- Usually sent differentially encoded as $\pi/4$ -DQPSK.
- Used in IS-136 (US digital cellular), PDC and PH5 (Japan), TETRA.



OQPSK time offset: $m_I(t)$ changes at even multiples of T , $m_Q(t)$ at odd multiples. Note the half-symbol offset. Rappaport Fig 5.30



$\pi/4$ -QPSK: constellation (a) and constellation (b) are used on alternate symbols, giving the 8-state pattern below with all allowed transitions drawn. Rappaport Fig 5.31

Why $\pi/4$ -QPSK is preferred over OQPSK 2

1. It can be detected non-coherently. Because it is differentially encoded, a differential or discriminator detector works. OQPSK must be detected coherently, and the half-symbol offset makes differential detection impossible.
2. Non-coherent detection $\rightarrow$ no carrier recovery circuit $\rightarrow$ far simpler, cheaper, faster-acquiring receiver.
3. Better in multipath spread and fading, where a coherent receiver cannot hold a phase reference. This is the mobile channel exactly.
4. Guaranteed transition every symbol $\rightarrow$ reliable symbol timing recovery even on long runs of identical dibits.
5. Cost accepted: envelope variation is larger than OQPSK's, so a mildly linear amplifier is still needed.

verified/added

4.4 Constant envelope modulation: BFSK, MSK, GMSK

Constant envelope (also called non-linear) means carrier amplitude never changes, whatever the data. **Why mobile radio loves it:** a class-C amplifier can be run into saturation at 60 to 70 % efficiency instead of a linear amplifier backed off to 20 %, which is the difference between a day and an afternoon of battery life. Also: very little out-of-band spillage, simpler receiver, and high immunity to random FM noise and Rayleigh fading.

BFSK, the starting point

- Two carrier frequencies f_{c1} (mark) and f_{c2} (space) carry 1 and 0.
- $s_i(t) = \sqrt{2E_b/T_b} \cos(2\pi f_{ci}t)$, $i = 1, 2$.
- Constant amplitude $\rightarrow$ constant envelope.
- Detected **coherently** (two correlators, compare outputs) or **non-coherently** (two BPFs + envelope detectors + comparator).
- (+) simple, insensitive to amplitude fluctuation, low P_e .
- (-) **needs more bandwidth than ASK or PSK**.
- (-) If the two oscillators are free-running the phase jumps at every transition $\rightarrow$ spectral widening, large sidelobes.
- Fix: **CPFSK**, one oscillator whose frequency is shifted, so phase stays continuous.

MSK = the best CPFSK

- **Definition 1 (as CPFSK):** a special case of CPFSK in which the peak frequency deviation is exactly **one quarter of the bit rate**, $\Delta f = R_b/4$, giving modulation index $h = 2\Delta f T_b = 0.5$.
- $h = 0.5$ is the **smallest index for which the two tones are orthogonal**, $f_1 - f_2 = 1/(2T_b)$. Hence “**minimum shift**”.
- Orthogonal $\rightarrow$ uncorrelated $\rightarrow$ the receiver separates them cleanly.
- **Definition 2 (as OQPSK):** MSK is OQPSK in which the rectangular data pulses are replaced by **half-cycle sinusoidal** pulses.
- That pulse shaping is what makes the phase change *linear* and bounded to $\pm\pi/2$ over a bit interval, so phase is continuous and smooth.
- Also called **Fast FSK**: the frequency shift is only half that of ordinary FSK.
- Good envelope, good spectral efficiency, good BER, no bandpass filter needed.

MSK as a special type of CPFSK and of OQPSK; compare OQPSK, MSK and GMSK; which suits GSM

2 4 3+3+2

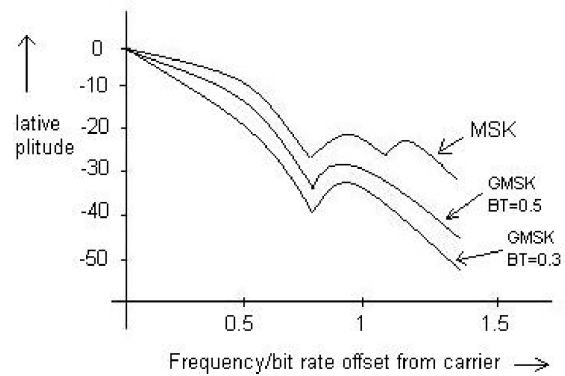
HOT 4/22 80 Ba, 81 Ba 3+3+2, 80 Ch 2, 76 Bh 4 · MSK and GMSK together are asked in 6 of 22 papers

	OQPSK	MSK	GMSK
Family	linear	constant envelope	constant envelope
Pulse shape	rectangular	half-cycle sine	Gaussian
Null-to-null BW	R_b	$1.5R_b$	below R_b
99 % power BW	narrowest of the three	$1.20R_b$	$0.86R_b$ at $BT = 0.25$
Sidelobe roll-off	f^{-2} , slowest	f^{-4}	fastest of the three
Envelope	near-constant	constant	constant
Power efficiency	$Q(\sqrt{2E_b/N_0})$	the same	slight loss from ISI
Amplifier needed	mildly linear	class C, saturated	class C, saturated
Detection	coherent only	coherent or not	FM discriminator works

Read the table as a story: OQPSK has the narrowest main lobe but the slowest-decaying tails, MSK trades a wider main lobe for tails that die four times faster, GMSK then squeezes the main lobe back down with a Gaussian filter and pays for it in ISI.

Which is suitable for GSM, and why 2

- **GMSK with $BT = 0.3$.** Three reasons, in order of weight:
 1. **Constant envelope** → the handset can use a saturated class-C power amplifier → battery life and cheap RF.
 2. **Compact spectrum** → fits the 200 kHz GSM carrier with acceptable adjacent channel interference. MSK alone leaks too much.
 3. **Detectable non-coherently** by a simple FM discriminator, so the receiver stays cheap.
- $BT = 0.3$ is the chosen compromise: tight enough spectrum, ISI penalty still small enough to be equalized out.



Occupied RF bandwidth

BT	90%	99%	99.9%	99.99%
0.2 GMSK	0.52	0.79	0.99	1.22
0.25GMSK	0.57	0.86	1.09	1.37
0.5 GMSK	0.69	1.04	1.33	2.08
MSK	0.78	1.20	2.76	6.00

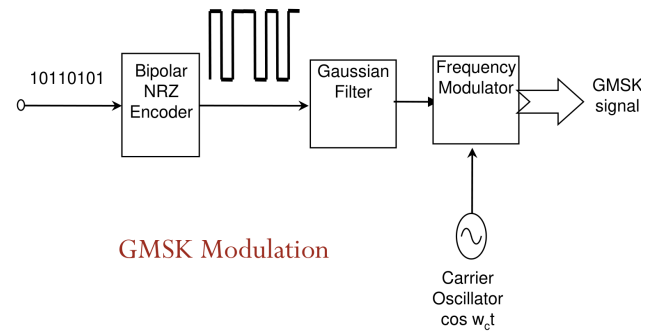
PSD of MSK against GMSK at three BT values, plus the occupied RF bandwidth table in units of R_b . **MSK's tails do not fall fast enough** to protect the adjacent channel, which is the entire reason GMSK exists. SST Ch 4

GMSK transmission and detection process, block diagram and constellation

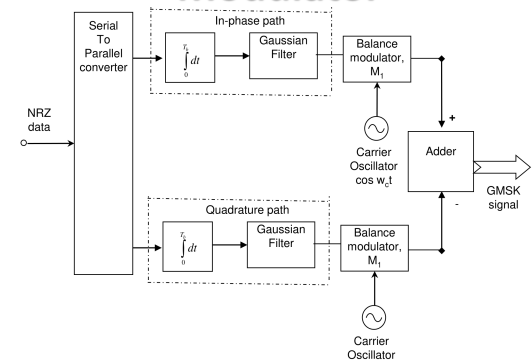
HOT 4/22 79 Bh **2+6**, 81 Ba **8**, 76 Bh **4**, 70 Ma **8** · 79 Bh pairs it with “advantages of digital modulation”, see §4.1

What GMSK is

- MSK preceded by a **premodulation Gaussian low-pass filter**.
- The Gaussian filter has **no zero crossings and no overshoot**, so it smooths the rectangular data into a shape that suppresses the high-frequency content responsible for MSK's sidelobes.
- Transfer function $H(f) = \exp\left[-\frac{\ln 2}{2} \left(\frac{f}{B}\right)^2\right]$, where B is the **3 dB bandwidth** of the filter.
- The filter is fully described by B and the symbol duration T , so GMSK is named by its **BT product**. $BT = \infty$ is plain MSK.
- **Smaller BT** : tighter spectrum, sidelobes fall away faster, but the pulse spreads over more symbols $\rightarrow$ more **ISI** $\rightarrow$ higher BER.
- GSM: $BT = 0.3$. CDPD and Mobitex also use GMSK.
- Constellation: same 4 points as QPSK, but the trajectory between them **follows the unit circle** instead of cutting across it, which is the constant envelope drawn.



GMSK transmitter, FM route. SST Ch 4



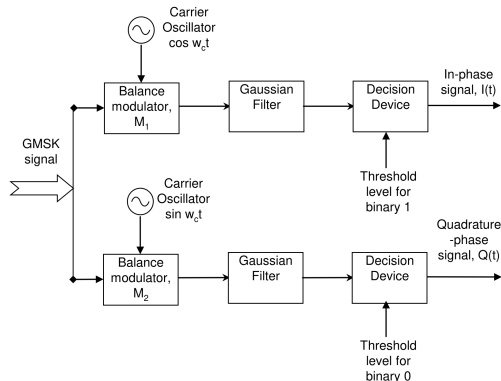
GMSK transmitter, I-Q route. SST, Fig 3.2.17

Transmission process 3

1. Data $\rightarrow$ **bipolar NRZ encoder** $\rightarrow \pm 1$ rectangular pulses.
2. $\rightarrow$ **Gaussian LPF** of bandwidth B , which rounds every pulse edge.
3. $\rightarrow$ **FM modulator** set to modulation index exactly 0.5, driven by the carrier oscillator $\cos \omega_c t$.
4. Output is GMSK: constant amplitude, continuous smoothly varying phase.
5. **I-Q alternative** (right-hand figure above): the filtered signal is integrated, split into in-phase and quadrature paths, each Gaussian-filtered and multiplied by $\cos \omega_c t / \sin \omega_c t$, then subtracted. **Advantage: the index stays at exactly 0.5 with no tuning or drift.**

Detection process 3

1. **Coherent (orthogonal) detector**, drawn left: GMSK splits into two arms.
2. Each arm has a **balance modulator** fed by $\cos \omega_c t$ or $\sin \omega_c t$, then a **Gaussian filter** matched to the transmitter's.
3. Each arm's **decision device** compares against its own threshold, giving $I(t)$ and $Q(t)$, which are recombined into the bit stream.
4. **Non-coherent alternative**: a simple **FM discriminator** followed by an integrate-and-dump. Much cheaper, small BER penalty. This is what a GSM handset actually does.



GMSK detector, orthogonal coherent form. SST, Fig 3.2.18

Worked example: sizing the Gaussian filter verified/added (tutorial problem, never a PYQ)**PROBLEM** SST tutorial · not from any paper

Find the 3-dB bandwidth for a Gaussian low pass filter used to produce 0.25 GMSK with a channel data rate of $R_b = 270$ kbps. What is the 90 % power bandwidth in the RF channel?

Step 1: symbol duration

- Binary GMSK, so one bit per symbol and
 $T = 1/R_b = 1/270,000 = 3.7 \mu\text{s}$.

Step 2: 3-dB bandwidth from the BT product

- “0.25 GMSK” means $BT = 0.25$, so
 $B = 0.25/T = 0.25/3.7 \mu\text{s} = 67.57 \text{ kHz}$.

Step 3: 90 % power RF bandwidth from the table

- Read the occupied RF bandwidth table (§4.4 figure): row $BT = 0.25$, column 90 % gives $0.57R_b$.
- RF BW = $0.57 \times 270,000 = 153.9 \text{ kHz}$.

Answer

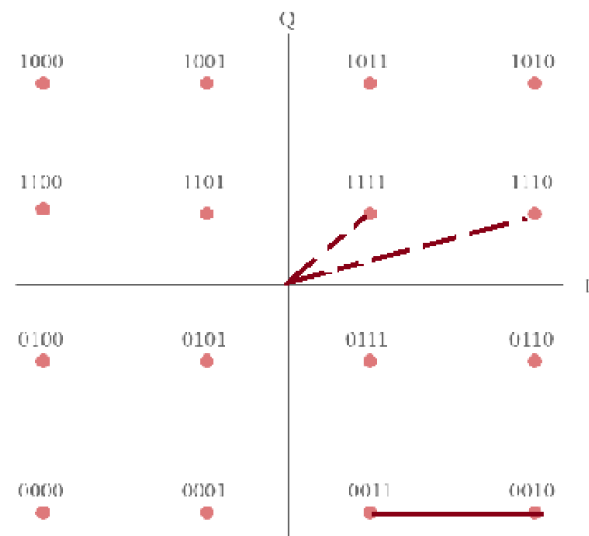
- 3-dB filter bandwidth $B = 67.57 \text{ kHz}$.
- 90 % power RF bandwidth = **153.9 kHz**.

Notes on this problem

- Real GSM uses $BT = 0.3$ and $R_b = 270.833 \text{ kbps}$** , not the rounded values here. Answer with whatever the paper prints.
- The table is in units of R_b , so always multiply the tabulated figure by the bit rate. Forgetting that is the standard slip.
- Sanity check: 153.9 kHz sits inside GSM’s 200 kHz carrier spacing, which is exactly the design intent.

4.5 M-ary modulation: MPSK, MFSK and QAM**The M-ary idea**

- Send one of $M = 2^n$ distinct waveforms per signalling interval, carrying n bits.
- Symbol duration $T_s = nT_b$, so the symbol rate drops by n for the same bit rate.
- Bandwidth needed falls by a factor of n** , i.e. η_B rises to n bps/Hz in principle.
- Cost: the M points crowd together, $d_{\min}$ shrinks, so **more power is needed** for the same BER. η_p falls.
- MPSK**: all M points on one circle of radius $\sqrt{E_s}$, differing only in phase. Constant envelope, but $d_{\min} = 2\sqrt{E_s} \sin(\pi/M)$ collapses as M grows.
- MFSK**: M orthogonal frequencies. Here bandwidth grows with M while power efficiency improves: the opposite trade to MPSK.
- QAM**: vary amplitude and phase, so points fill a 2-D grid instead of a circle.
- 3G and later lean on these: EV-DO adapts between QPSK, 8-PSK and 16-QAM by channel quality.



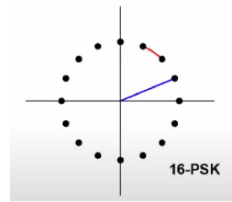
16-QAM, a 4×4 grid, 4 bits per symbol. Nearest-neighbour distance is the same everywhere, which is why the grid beats the circle. SST Ch 4

Why do we need M-ary QAM? Compare 16-QAM and 16-PSK using constellation diagrams 2+6

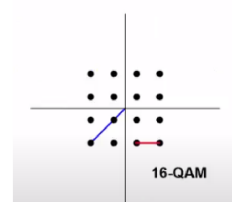
PIN 1/22 77 Ch 2+6

Why M-ary QAM is needed 2

- Spectrum is licensed and finite, data demand is not. Binary schemes cap out near 1 bps/Hz.
- M-ary raises η_B to $\log_2 M$ bps/Hz without extra bandwidth.
- But **M-ary PSK alone fails at high M** : all points sit on one circle, so $d_{\min}$ falls like $\sin(\pi/M)$ and the power cost explodes.
- **QAM uses the radial dimension too**. Spreading points over a 2-D grid gives a much larger $d_{\min}$ for the same average power.
- So M-ary QAM is the only practical way to reach 4, 6 or 8 bits per symbol on a band-limited channel.



16-PSK (left): 16 points on one circle, all at the same power, spacing 22.5° .



16-QAM (right): 16 points on a 4×4 grid, three different amplitudes. Same 4 bits per symbol either way. SST Ch 4

Parameter	16-PSK	16-QAM
Bits per symbol	4	4
Bandwidth	same	same (identical symbol rate)
Constellation	16 points, one circle	4×4 grid, 3 amplitude levels
Envelope	constant	varies
Amplifier	non-linear, efficient	linear, backed off
$d_{\min}$ at equal peak power	smaller	larger
Power for the same $d_{\min}$	1.6 dB more peak power	reference
Power at equal $d_{\min}$ (average)	4.19 dB more average power verified/added	reference
Noise immunity	equal, if $d_{\min}$ is equalised	equal, at less power

The number the examiner wants

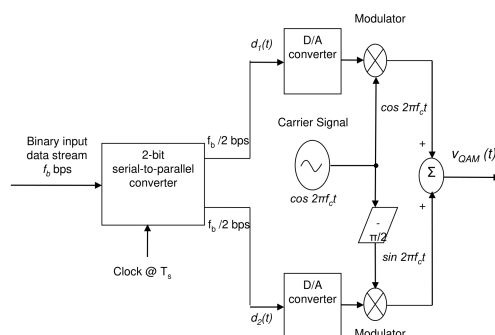
- **At the same minimum Euclidean distance (the same noise immunity), 16-QAM needs 1.6 dB less peak power than 16-PSK.**
- Measured on *average* power instead, the gap is **4.19 dB** verified/added.
- Both figures describe the same fact: the grid packs 16 points more economically than the circle.
- **Careful:** the lecture note's line "noise immunity is same" is only true when the two are compared at equal $d_{\min}$. At equal transmit power 16-QAM is clearly better.

So why is 16-PSK used at all

- Constant envelope $\rightarrow$ the transmitter can saturate its amplifier.
- 16-QAM's amplitude variation forces a linear amplifier with 6 to 8 dB of back-off, which **gives back much of the 4.19 dB**.
- Rule of thumb: QAM on the downlink (base station has mains power), constant envelope on the uplink (handset does not).
- Both need coherent detection and accurate gain control.

QAM modulator, walked through

1. Binary stream at f_b enters a **2-bit serial-to-parallel converter** (4-bit for 16-QAM), clocked at the symbol rate T_s .
2. Each half feeds a **D/A converter** that turns the bit group into a multi-level signal $d_1(t)$ and $d_2(t)$.
3. $d_1(t)$ multiplies $\cos 2\pi f_c t$, $d_2(t)$ multiplies $\sin 2\pi f_c t$ (carrier through a $\pi/2$ shifter).
4. Summer adds the two $\rightarrow$ a signal whose amplitude and phase both carry data.
5. **Demodulator** is the QPSK receiver with A/D converters instead of 1-bit decision devices, plus a carrier recovery circuit (raise to 4th power, BPF at $4f_c$, divide by 4).



QAM modulator. SST, Fig 3.2.21

4.6 OFDM the single most examined topic in this chapter

Orthogonal frequency division multiplexing. One high-rate carrier replaced by many low-rate ones, spaced so they overlap without interfering. It is the air interface of WiFi, WiMAX, LTE and 5G, and it is asked in **9 of the 22 papers** here, always with a block diagram attached.

OFDM operation, block diagram of transmitter and receiver 4 8

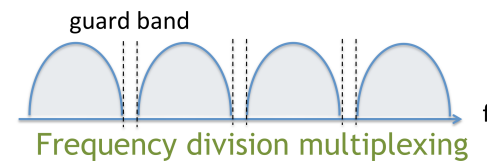
TOP 9/22 74 Bh 8, 80 Bh 8, 78 Ch 8, 71 Ma 8, 70 Bh 4, 79 Ch 4 (Tx only), 80 Ch 4 (Rx only), 82 Bh 4+4, 70 Ma 8 · **9 of 22 papers.**
Nothing else in the chapter comes close.

The problem OFDM solves

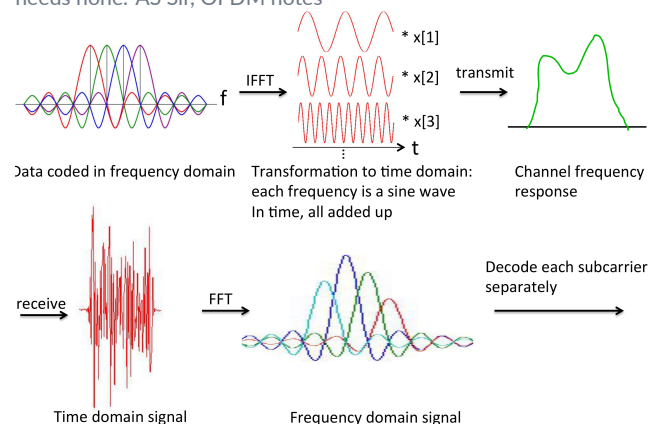
- Push the bit rate up on a **single carrier** and the symbol period T_s shrinks.
- Once T_s approaches the channel's **delay spread** σ_τ , echoes of symbol n land on top of symbol $n+1$ → severe **ISI** and frequency-selective fading.
- Fixing that in the time domain needs an equalizer with a great many taps, which is expensive and must adapt continuously.
- **OFDM's answer: do not make the symbol shorter, make it longer, and send many of them side by side.**

The principle 4

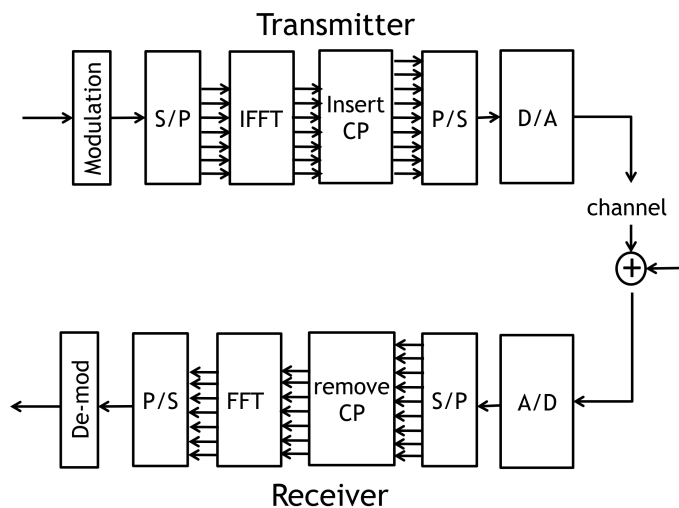
- Split the high-rate stream into N **parallel low-rate substreams**.
- Each substream modulates its own **subcarrier**, using QPSK or QAM.
- Each subcarrier's symbol period becomes NT_s , now **much longer than the delay spread** → each subchannel sees **flat fading**, not selective.
- Flat fading → one complex multiply per subcarrier undoes the channel. **The many-tap equalizer disappears.**
- Subcarrier spacing is set to $\Delta f = 1/T$, which makes them **orthogonal**: each one's spectrum has a null exactly where every other one peaks.
- So the spectra may **overlap without interfering**, and no guard bands are needed. Against FDM that is roughly a **50 % bandwidth saving**.
- The whole bank of N modulators is computed as one **IFFT**, and the bank of demodulators as one **FFT**. No oscillator bank, no filter bank.



The figure that answers "OFDM principle". FDM (top) wastes spectrum on guard bands. OFDM (bottom) overlaps orthogonal subcarriers and needs none. AS Sir, OFDM notes



Data is coded in the **frequency** domain, IFFT turns it into a time waveform, the channel acts, FFT recovers each subcarrier separately. AS Sir, OFDM notes



Draw exactly this for the 8-mark question. Transmitter on top, receiver below, channel and noise between them. AS Sir, OFDM notes

Cyclic prefix, the detail that earns marks

- Even with long symbols, the tail of one OFDM symbol still smears into the next.
- A blank guard interval would work, but the hardware must transmit continuously and the exact delay spread is unknown.
- **Solution: copy the last T_g of the symbol and glue it on the front.**
- Two effects:
 - Any echo arriving within T_g falls inside the copied portion, so it corrupts only the prefix, which is thrown away → **ISI removed**.
 - Linear convolution with the channel becomes *circular* convolution, so after the FFT each subcarrier is simply scaled: $Y[k] = H[k]X[k] \rightarrow$ **one-tap equalization**.
- Requirement: $T_g >$ maximum delay spread. In 802.11, CP : data = 1 : 4.
- Side benefit: the packet still decodes even if detected a little late.
- **Cost: the CP is pure overhead**, up to 20 to 25 % of throughput.

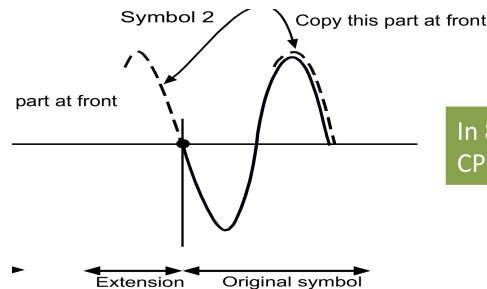
Transmitter chain 4

1. **Modulation / symbol mapping:** bits $\rightarrow$ QPSK or QAM symbols (after channel coding and interleaving).
2. **S/P:** serial symbol stream split into N parallel subcarrier values, i.e. a frequency-domain block.
3. **IFFT:** converts that block to N time-domain samples. This is the modulation.
4. **P/S:** back to a serial sample stream.
5. **Insert CP:** cyclic prefix prepended (see below).
6. **D/A** $\rightarrow$ up-convert to RF $\rightarrow$ antenna.

Receiver chain 4

1. Down-convert $\rightarrow$ **A/D**.
2. **Synchronise:** packet detection, then carrier frequency offset correction.
3. **Remove CP**, discard the guard samples.
4. **S/P** $\rightarrow$ block of N time samples.
5. **FFT** $\rightarrow$ back to the frequency domain, one value per subcarrier.
6. **Equalize:** divide each subcarrier by its estimated $H[k]$, one tap each.
7. **P/S** $\rightarrow$ **de-modulation / de-mapping** $\rightarrow$ deinterleave and decode $\rightarrow$ data out.

Receiver is the transmitter read right to left. Say that and you have drawn half the answer.



In 802.11,
CP: data = 1:4

The tail of the symbol is copied to the front as an extension. AS Sir, OFDM notes

Two more practical details

- **Virtual (null) subcarriers** at the band edges are left unused: edge bins are the most vulnerable to frequency offset, and the gap suppresses out-of-band radiation.
- **Pilot subcarriers** carry known symbols so the receiver can track residual phase and estimate $H[k]$.
- 802.11a splits its 64 bins as **48 data + 4 pilot + 12 null** **verified/added**. The lecture note's "48 of 64 occupied" counts only the data bins.

OFDM: advantages and disadvantages 4

PIN 1/22 82 Bh 4+4 · asked together with “draw the transmitting side”, so budget the time

Advantages

- **High spectral efficiency:** overlapping orthogonal subcarriers, no guard bands, roughly twice FDM's.
- **Immune to ISI and frequency-selective fading:** long symbols plus the CP.
- **Simple equalization:** one complex multiply per subcarrier, no adaptive transversal filter.
- **Efficient to build:** FFT/IFFT in DSP, cost grows as $N \log N$, not N^2 .
- **Robust to narrowband interference:** a jammer kills a few subcarriers, and coding + interleaving recovers them.
- **Flexible:** power and modulation order can be set per subcarrier (adaptive bit loading); extends naturally to OFDMA and MIMO.

Disadvantages

- **High PAPR (peak-to-average power ratio):** N subcarriers can add in phase, so the amplifier needs heavy back-off → poor efficiency. **The main drawback.** It is why LTE uses SC-FDMA, not OFDMA, on the uplink.
- **Very sensitive to frequency offset and phase noise:** any error destroys orthogonality → inter-carrier interference (ICI).
- **Sensitive to Doppler:** a fast-moving mobile changes the channel within one long symbol.
- **CP overhead** wastes 20 to 25 % of both time and energy.
- **Tight synchronisation** needed in both time and frequency.

Used in: DAB, DVB-T, 802.11a/g/n/ac, WiMAX, LTE downlink, 5G NR, ADSL.

4.7 Spread spectrum: PN sequences, DSSS and FHSS

Define spread spectrum modulation; DSSS with transmitter and receiver block diagrams 5 6 2+6

HOT 4/22 82 Bh 2+6, 81 Ba 6, 80 Ch 2+5, 73 Ma 4 · spread spectrum overall: 7 of 22 papers

Definition 2

- A modulation in which the transmitted signal occupies a bandwidth **far greater than the minimum needed to send the message.**
- Spreading is done by a **code independent of the data**, usually a PN sequence.
- The receiver **de-spreads** using a synchronised copy of the same code.
- **Only a receiver holding the code can recover the message.** To anyone else the signal looks like a rise in the noise floor.
- **Processing gain** $PG = \frac{B_{ss}}{B} = \frac{T_b}{T_c} = N$ chips per bit. This number is the jamming margin.

The two problems it answers

- How to secure information over a broadcast medium.
- How to survive deliberate jamming and external interference.
- (And, as a bonus, how to let many users share one band with no coordination.)

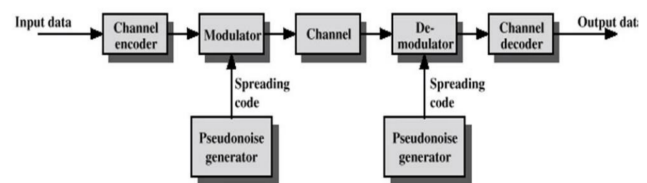
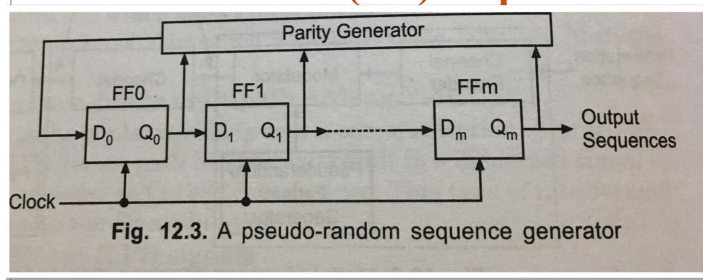


Figure 7.1 General Model of Spread Spectrum Digital Communication System

General model of a spread spectrum system. Note the symmetry: the same pseudonoise generator appears at both ends, and it sits outside the channel coding. SST Ch 4, Fig 7.1



PN sequence generator: m D flip-flops in a shift register, XOR parity feedback into D_0 . Maximal length $2^m - 1$. SST Ch 4, Fig 12.3

DSSS transmitter 3

1. Original data bit stream at rate R_b , in ± 1 form.
2. **Chip (PN) generator** produces a code at rate $R_c = NR_b$, i.e. N chips per data bit.
3. **Multiplier** multiplies data by chips. **Each data bit is replaced by N chips**, which is where the bandwidth expansion happens.
4. The wideband product PSK-modulates the carrier $\rightarrow$ transmitted DSSS signal.
5. Spectrum is now N times wider and N times lower in power density, so the signal can sit *beneath* the noise floor.

DSSS receiver 3

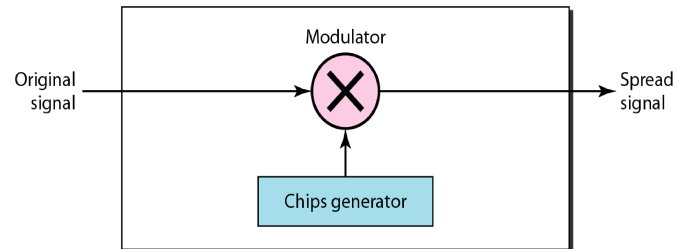
1. BPF and PSK demodulation give back the wideband chip stream.
2. A **local PN generator**, synchronised to the transmitter's, produces the same code.
3. **Multiply again by the same code**: since $(\pm 1)^2 = +1$, the data collapses back to its original narrow bandwidth (de-spreading).
4. **Anything not multiplied by the matching code gets spread instead of de-spread**, so interference and other users are pushed down by the processing gain.
5. Narrowband filter + integrate-and-dump $\rightarrow$ recovered data.

DSSS (+)

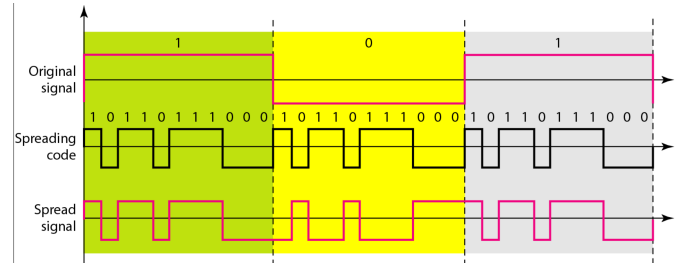
- Multiple access with no coordination $\rightarrow$ **CDMA**, IS-95 and 3G.
- Narrowband jamming is suppressed by the full processing gain.
- Multipath components arriving more than one chip apart are resolvable $\rightarrow$ **RAKE** receiver turns them into diversity (Ch 5).
- No frequency planning: every cell can use every frequency.

PN sequence properties 2

- Coded sequence of 1s and 0s with **noise-like** statistics but deterministic generation.
- **Balance**: 1s and 0s occur with near-equal probability.
- **Shift property**: adding a shifted copy of a PN sequence gives back the same sequence in a different phase.
- **High autocorrelation** at zero lag, near zero elsewhere $\rightarrow$ the receiver can find the exact code phase.
- **Low cross-correlation** with other codes $\rightarrow$ other users look like noise.
- Easy to generate and to synchronise, in hardware, at chip rate.



DSSS spreading. SST Ch 4



DSSS waveform example: one data bit, an 11-chip code, and their product. Data 1 passes the code through, data 0 inverts it. SST Ch 4

DSSS (-)

- **Near-far problem**: a close transmitter swamps a distant one, so tight and fast power control is mandatory.
- PN acquisition and tracking is slow and hardware-hungry.
- **Self-jamming**: code cross-correlation is low but not zero, so users interfere with each other as the load rises.

FHSS operation with a neat block diagram 4 5

HOT 3/22 79 Ch 2+5, 73 Bh 4, 73 Ma 4 · as FHMA this is a TOP-tier Ch 7 question, see §7.3

Operation 4

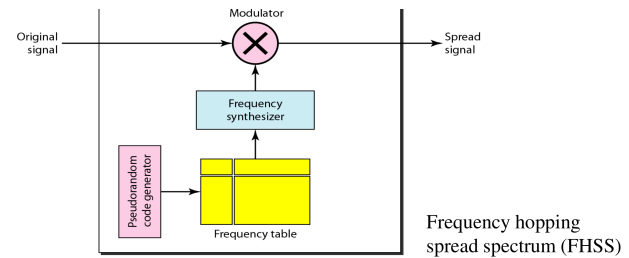
- FHSS uses M different carrier frequencies, and the carrier hops among them in a pseudo-random pattern.
- Binary data is applied to an **M-ary FSK modulator**.
- A **pseudorandom code generator** drives a **frequency synthesizer**, which produces the whole range of hop frequencies from one reference.
- The synthesizer output at any instant is the **frequency hop**, chosen by the current k code bits from a frequency table of 2^k entries.
- A **mixer** combines each hop with the MFSK signal → transmitted signal.
- **The spectrum is spread sequentially, not instantaneously:** at any one moment the signal is narrowband, but over time it covers the whole band.
- The receiver runs the same code and synthesizer, so its local oscillator hops in lockstep and de-hops the signal back to a fixed IF.

Slow against fast hopping

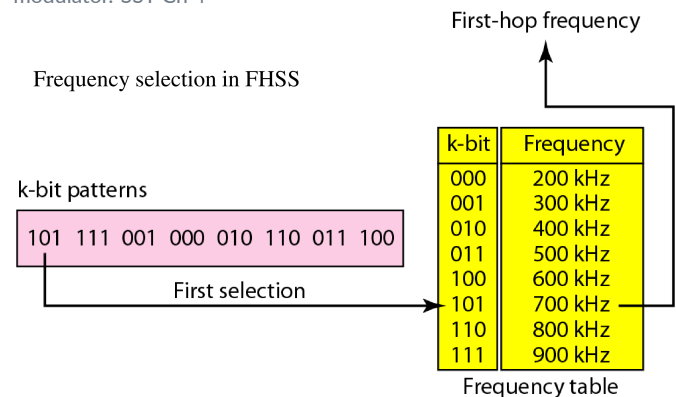
- **Slow FHSS:** several symbols per hop ($R_h < R_s$). Cheaper synthesizer, used by GSM and Bluetooth.
- **Fast FHSS:** several hops per symbol ($R_h > R_s$). Each symbol gets frequency diversity, so a bad hop cannot kill it. Costs a fast synthesizer.

FHSS (+)

- **Immune to the near-far problem:** two users are almost never on the same frequency at the same instant, so no tight power control is needed.
- Resists partial-band jamming and frequency-selective fading, because the signal moves out of the bad band on the next hop.
- Synchronisation is easier and faster than DSSS's chip-level acquisition.
- Longer range for the same power, since instantaneous bandwidth stays narrow.



FHSS transmitter. Code generator → frequency table → synthesizer → modulator. SST Ch 4



Frequency selection: each 3-bit group of the PN stream indexes one row of the frequency table, which is the next hop. SST Ch 4

FHSS (-)

- Needs a fast, agile and expensive **frequency synthesizer**.
- **Hits:** when two users pick the same hop, both symbols are lost, so error control coding is mandatory.
- Detection is usually non-coherent (the phase is lost at every hop), costing about 3 dB.
- Total bandwidth is large, so it must be licensed or confined to an ISM band.

Bluetooth: 79 hop channels, 1600 hops/s, 625 μ s dwell.

Advantages and major disadvantages of spread spectrum 2 4

76 Bh 4 (+), 77 Ch 2 (-)

Advantages 4

- **Anti-jam:** a narrowband jammer is spread by PG at the receiver, so its effective power drops by the processing gain.
- **Low probability of intercept:** the signal sits under the noise floor, so an eavesdropper cannot even tell it is there.
- **Privacy:** without the code the signal cannot be de-spread.
- **Multiple access:** near-orthogonal codes let many users occupy one band at once (CDMA).
- **Multipath rejection:** delayed copies decorrelate with the code and are either rejected or harvested by a RAKE receiver.
- **Graceful degradation:** adding users raises the noise floor a little rather than blocking calls outright.
- Mitigates ISI, and **bandwidth efficiency actually rises as simultaneous users increase.**

Major disadvantages 2

- **Large bandwidth** needed, many times the message bandwidth. Spectrum is expensive.
- Complex and costly transmitter and receiver: PN generation, acquisition, tracking.
- **Long acquisition time** before the first bit can be decoded.
- **Near-far problem** in DSSS → fast closed-loop power control needed.
- **Self-jamming** from non-zero code cross-correlation, which caps capacity.
- Synchronisation of the two PN generators must be exact to within a fraction of a chip.

4.8 Performance of modulation in fading channels syllabus, no PYQ yet

What the fading channel does to BER

- In **AWGN** the error probability falls *exponentially* with E_b/N_0 .
- In **flat Rayleigh fading** it falls only *inversely*: $\bar{P}_e \approx 1/(4 E_b/N_0)$ for coherent BPSK.
- Consequence: reaching 10^{-3} BER needs about **7 dB in AWGN and about 24 dB in Rayleigh fading**. Tens of dB of extra power for the same quality. verified/added
- Cause: the deep fades dominate the average, not the typical signal level.
- Cure is **diversity**, not more power (Ch 5).

Which scheme survives the mobile channel

- **Coherent** schemes suffer most: the receiver cannot track a phase reference that is being randomised by the channel.
- **Differential and non-coherent** schemes (DPSK, $\pi/4$ -DQPSK, GMSK with a discriminator) degrade far more gracefully.
- **Constant envelope** schemes suit the handset, because the power amplifier can saturate.
- **Linear** schemes (QAM) need back-off and accurate gain control, so they belong on the downlink or on a fixed link.

The irreducible error floor

- In **frequency-selective** fading, ISI produces errors that **no amount of transmit power removes**.
- The floor is set by the ratio σ_τ/T_s , i.e. delay spread against symbol period.
- In **fast fading**, random FM from Doppler produces a second floor set by $f_d T_s$.
- Four ways out: equalization, spread spectrum + RAKE, OFDM (long symbols), and diversity.

The trade-off table in one line each

- **BPSK:** most power efficient binary scheme, poorest η_B .
- **QPSK / OQPSK:** double η_B at no BER cost, envelope problems.
- **MSK / GMSK:** constant envelope, non-coherent detection, GSM's choice.
- **M-QAM:** highest η_B , needs high SNR and a linear amplifier.
- **DSSS:** trades a huge bandwidth for interference immunity and multiple access.
- **OFDM:** turns a selective channel into many flat ones, pays in PAPR.

4.9 Last-minute recall for this chapter

Must memorise

- QPSK: 2 bits/symbol, $T_s = 2T_b$, null-null BW = R_b , **same BER as BPSK**.
- MSK: CPFSK with $h = 0.5$, $\Delta f = R_b/4$, equals OQPSK with half-sine pulses.
- GMSK: MSK + Gaussian LPF, $H(f) = \exp[-\frac{\ln 2}{2}(f/B)^2]$, **GSM uses $BT = 0.3$** .
- Smaller $BT \rightarrow$ tighter spectrum, more ISI.
- OFDM: $\Delta f = 1/T$, IFFT at the transmitter, FFT at the receiver, CP > delay spread, main drawback **PAPR**.
- Spread spectrum: $PG = B_{ss}/B = T_b/T_c = N$.
- PN generator: m stages, maximal length $2^m - 1$.
- 16-QAM against 16-PSK: **1.6 dB less peak power** at equal $d_{\min}$, 4.19 dB less average power.
- $\pi/4$ -QPSK: max phase change 135° , differentially detectable, IS-136 and PDC.

No paper in 2070–2082 has set a numerical from this chapter. The one calculation worth holding is the Gaussian-filter sizing in §4.4, which is a tutorial problem, not a PYQ. Everything examined here is descriptive, and the marks live in the figures.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Most repeated, in order

1. OFDM transmitter and receiver +Fig **TOP 9/22**
2. Spread spectrum, DSSS and FHSS +Fig **TOP 7/22**
3. MSK and GMSK, and why GSM chose GMSK **TOP 6/22**
4. QPSK transmission and detection +Fig **HOT 4/22**
5. OFDM (+) and (-) **PIN 1/22**
6. 16-QAM against 16-PSK **PIN 1/22**
7. DPSK transmitter and receiver, PN sequence **PIN 1/22**
8. OQPSK, and why $\pi/4$ -QPSK beats it **PIN 1/22**

Draw first, then annotate. An unlabelled correct block diagram still scores; three paragraphs with no diagram do not.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB,	double, single, vestigial sideband
		VSF	
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	BT	GMSK filter bandwidth $\times$ symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
WLAN	wireless local area network	WMAN	wireless metropolitan area network
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
5G	5th generation mobile communication network	5G	5th generation mobile communication network

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.